

Saraf Anzum Shreya

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RESEARCH INTERESTS

Deep learning and computer vision for problems with real diagnostic and human impact. My work spans medical image analysis, including brain tumor classification and blood cell analysis, as well as assistive machine learning for visually impaired users. I am particularly interested in trustworthy AI where model outputs support real-world decisions. Across my projects, I focus on building end-to-end systems from data and model design to usable applications.

ACADEMICS

B.Sc. in Electronics & Telecommunication Engineering *Mar 2022 – Aug 2026*

Rajshahi University of Engineering & Technology (RUET), Rajshahi, Bangladesh

CGPA: 3.50 / 4.00 | **Best SGPA:** 3.83 / 4.00 (6th semester)

Relevant coursework: Machine Learning, Neural Network & Fuzzy Systems, Data Scienc, Digital Image Processing, Data Structure and Algorithm, Numerical Methods in Engineering, Digital Signal Processing, Linear Algebra and Three Dimensional Geometry, and Complex Variable and Statistical Analysis

PUBLICATIONS

Published Articles

- **Shreya, S.**, et al. (2024). A Hybrid Approach for Accurate Brain Tumor Detection Using Deep Learning Techniques. *Conference Paper, 1st author*.
doi:[10.1109/ICCIT64611.2024.11022592](https://doi.org/10.1109/ICCIT64611.2024.11022592)
- **Shreya, S.**, et al. (2024). Residual Block-Driven CNN for Accurate White Blood Cell Image Analysis and Classification. *Conference Paper, 4th author*.
doi:[10.1109/ICCIT64611.2024.11021843](https://doi.org/10.1109/ICCIT64611.2024.11021843)
- **Shreya, S.**, et al. (2022). Women Empowerment and Governance through Digitization of NGO Management Systems. *Conference Paper, 3rd author*.
doi:[10.1109/ICRPSET57982.2022.10188564](https://doi.org/10.1109/ICRPSET57982.2022.10188564)

Pre-prints

- **Shreya, S.**, et al. (2025). DB-FGA-Net: Dual Backbone Frequency Gated Attention Network for Multi-Class Brain Tumor Classification with Grad-CAM Interpretability. *Undergraduate Thesis, 1st author*.
doi:[10.48550/arXiv.2510.20299](https://doi.org/10.48550/arXiv.2510.20299)
- **Shreya, S.**, et al. (2025). Real-Time Currency Detection and Voice Feedback for Visually Impaired Individuals. *1st author*.
doi:[10.48550/arXiv.2510.20267](https://doi.org/10.48550/arXiv.2510.20267)

PROFESSIONAL EXPERIENCE

Industrial Trainee — BrainStation-23

March 2025

Data Scraping and Web Development

- Completed a two-week software-based attachment focused on web scraping and backend development.
- Developed a fully functional web data scraper using BeautifulSoup, Selenium, HTML, CSS and Flask to extract and display dynamic web content in a user-friendly interface.
- Project repository:
github.com/SarafAnzumShreya/Scraper

Independent Artist / Graphic Designer

- Create and design unique merchandise on a global print-on-demand marketplace.
- Shop:
[redbubble.com/people/taylor-shreya/shop](https://www.redbubble.com/people/taylor-shreya/shop)

TECHNICAL SKILLS

Programming: Python · C · C++ · HTML · CSS · JavaScript

Machine learning & vision: TensorFlow · Keras · Scikit-learn · OpenCV · YOLO · CNN

Databases: MySQL

Tools: Canva · AutoCAD · MATLAB · Mathematica · Arduino · ESP32-CAM · Flask

SELECTED PROJECTS

- **EchoCash** (Python, HTML, CSS, JavaScript, Flask, Socket.IO) — real-time currency recognition with voice feedback, announcing the detected currency and running total to support financial independence.
GitHub: github.com/SarafAnzumShreya/EchoCash
- **Avatar Me** (HTML, CSS, JavaScript) — web app for creating customized avatars (clothing, accessories, hairstyles, facial hair, hijabs) with inclusive options across different races and genders.
sarafanzumshreya.github.io/Avatar-Me
- **Reverse Vending Machine** (C, AtMega328p, ESP32-CAM) — ML-based bottle recognition with a secure deposit area; scans a QR code to credit the user's account, redeemable as usable currency, with bin-full alerting.
GitHub: github.com/SarafAnzumShreya/RVM

AWARDS & LEADERSHIP

- **Technical Scholarship**, Rajshahi University of Engineering & Technology, 2023–2026.
- **Huawei Seeds For The Future 2025** — Finalist; completed a 40-hour program on 5G, AI, Cloud Computing and Digital Power, one of 21 selected from 3,500+ applicants.
- **Cognizant 2.0 Case Competition** — 1st Runner-up out of 57 teams, for a data-driven solution to fire hazards in Bangladesh's garment industry, 2023.
- **Line Following Robot (LFR) Competition** — Participant, 'ETechnovation-2022', RUET, 2022.
- **Volunteer, ETechnovation** — departmental technical event at RUET, 2022.
- **Competitive Programming** — participated in multiple contests, individually and in teams, 2022–2023.
- **Social Media Community Management** — grew entertainment-focused communities on Instagram and X/Twitter to 10k–13k followers, 2019–2021.

CERTIFICATIONS

- **Advanced Learning Algorithms** — DeepLearning.AI, Stanford University, 2025. Credential ID: [6T7KI3YQGJN0](#)
- **Supervised Machine Learning: Regression and Classification** — DeepLearning.AI, Stanford University, 2024. Credential ID: [Y9815S6QCI3W](#)